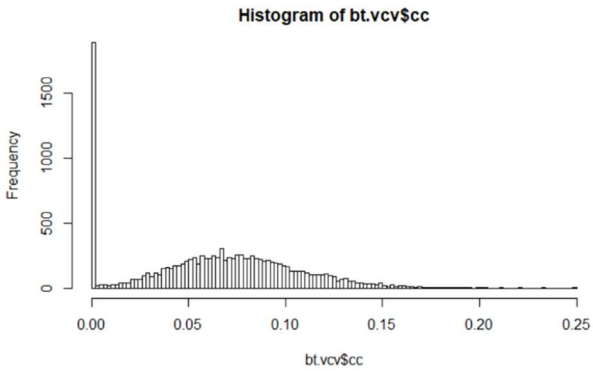
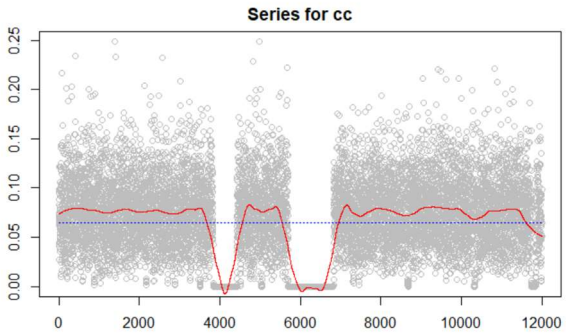
AN ILLUSTRATIVE CHAIN FOR THE BLUE TIT DATA

Every run of an MCMC chain is different. In one of the runs developing the long chain of 10,000 samples I had the following interesting example of the results for 'fosternest', but I cannot capture them in an html as to do so would re-run the chain!

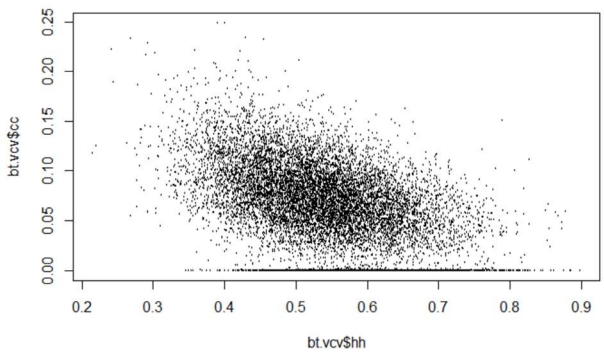
The 2 plots below show the histogram for c^2 and its series.

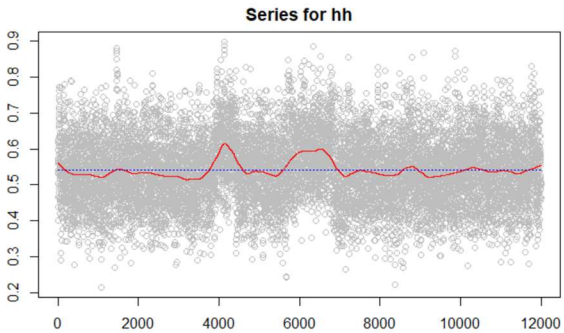




These demonstrate one of the issues with MCMC methods for estimating variances. The standard priors for a variance will often be strongly shaped like a reversed J, which keeps the estimate away from 0 but allows it to be arbitrarily close. With these priors, once a very low value for the variance is sampled it is often difficult to move away from the vicinity of 0 – the 0 can act like a black hole. The series plot shows this behaviour. The histogram shows a spike and the posterior $Prob(c^2 < 0.001) = 0.155$. The series plot shows a short chain may or may not display a spike of varying size.

The estimates from the means of the posterior distributions were $c^2 = 0.065$ with s.d. 0.041, and $h^2 = 0.542$ with s.d. 0.094. The 'fosternest' is a source of covariance among sibs, and if the variance attributed to 'fosternest' is close to 0, some of the covariance is interpreted as additive variance, i.e. h^2. The plots below show the joint distribution of c^2 and h^2, which has a correlation of -0.469; and the series for h^2, which shows peaks where c^2 is persistently low.





The conclusion is true convergence may take a long while to account fully for this behaviour.